Reg. No. | | | | | | | | | | |

[14EE64]

SIXTH SEMESTER B. Tech EXAMINATION, JUNE–2019

ELECTRICAL DRAWING

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full questions choosing atleast **TWO** full from each part
2. Missing data may be suitably assumed.

PART - A

1. Draw the single line diagram of a typical substation having following data: **14**
   - **i.** incoming line :11KV,two
   - **ii.** outgoing line : 33KV,four
   - **iii.** Transformer :11/33 KV,star/delta connected ,two
   - **iv.** Buses :11KV,one
     33 KV,two

   Show the positions of other necessary electrical equipment's like bus coupler, lightning arrestor , P.T.

2. Draw the developed winding diagram of an armature of a DC machine with the following details: **14**
   Number of poles =4. Number of slots =24, number of coil sides per slot =2 .Type of winding is simplex lap progressive. Draw the winding table & sequence diagram.

3. Draw the developed winding diagram for an AC motor with following data: **14**
   - **i.** number of slots =12
   - **ii.** number of poles =4
   - **iii.** number of phases =3.

   Winding type : double layer ,full pitched wave winding with star connection.

4. Draw a neat schematic diagram of a thermal power plant. **14**

PART - B

5. Following are the details of a main pole of a DC machine . Draw to suitable scale **14**
   Sectional Elevation &
   End view
   Number of poles =4
   Height of pole =190mm
   Width of the pole =140mm
   Length of the pole =200mm
   Armature diameter = 480mm
   Pole arc/ pole pitch =.67
   Number of turns per pole =1890
   Conductor area =2mm^2
   Depth of the winding =40mm
   Height of the winding =140mm

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6. The sketch number 6 shows the details of commutator parts. Draw the following view : **14**

i. End elevation

ii. Front elevation . Commutator segments are 70 in number, with 1.5 mm thick insulation provided in between.

7. Draw to a suitable scale **14**

i. the half sectional elevation and

ii. Half sectional plan of a single phase shell type power transformer with the following details:

Core:

Cross section of the core leg for the total flux =14cm x 37 cm

Height of stack of laminations = 37cm

Window :13 cm wide x 24cm long

Overall dimension of the core:

Width =54cm; length =38 cm ; height = 37 cm

LV winding:

Number of turns = 40

Number of sections = 4

Turns per section =10

Overall dimensions of each sections =1.7 cm x 11 cm

HV winding:

Number of turns = 180

Number of sections =4

Turns per section =45

Overall dimensions of each section = 1.7cmx 11 cm

The HV & LV coils are to be arranged for medium reactance.

Insulation details: Insulation between end coils and iron =1cm

Spacers between LV & HV sections =1.7 cm

Spacers between LV sections = 0.3cm

Spacers between HV sections = 0.6 cm

8. The sketch no 8 shows the details of stator of an alternator. Draw to suitable scale **14**

i. End view right half in section

ii. Elevation top half in section.

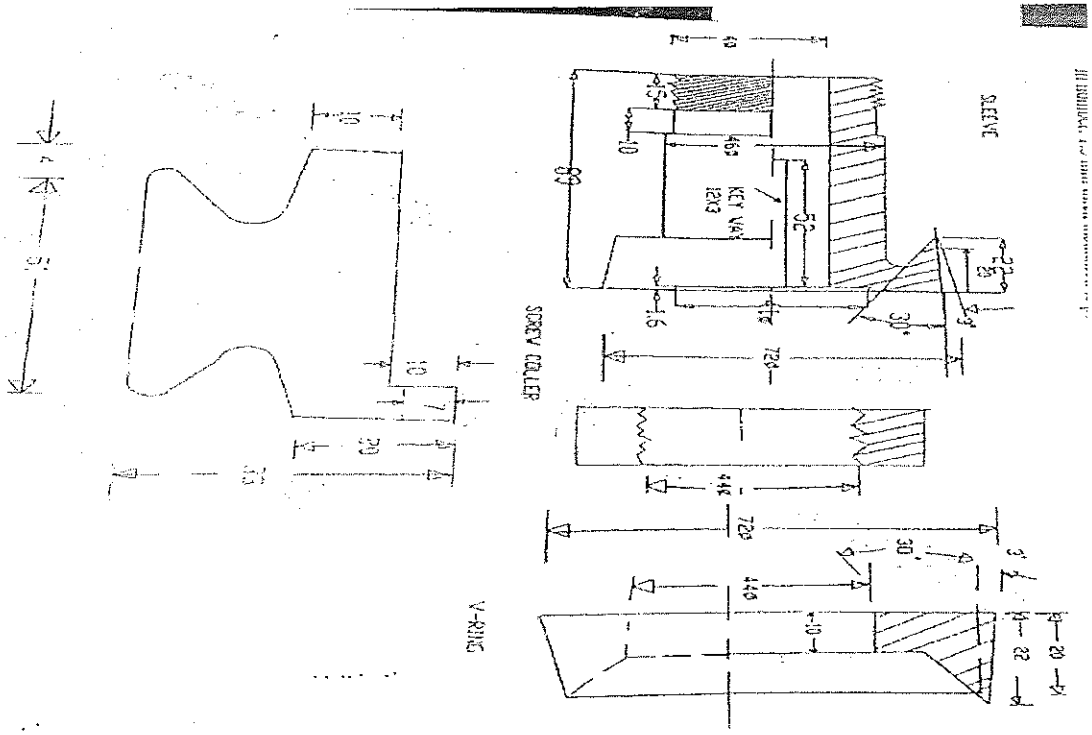
SLEEVE
SCREW COLLER
V-RING

Fig-6.

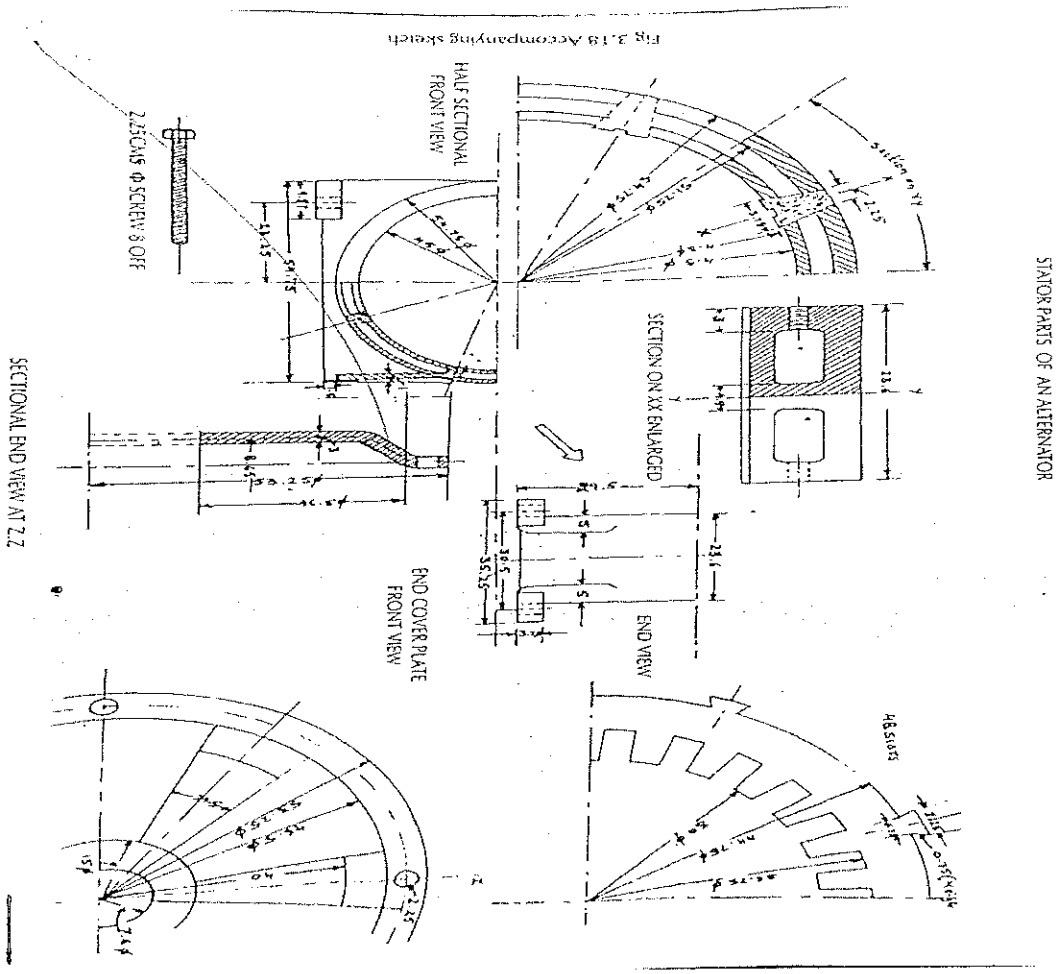
STATOR PARTS OF AN ALTERNATOR
SECTION ON XX ENLARGED
END VIEW
HALF SECTIONAL FRONT VIEW
END COVER PLATE FRONT VIEW
SECTIONAL END VIEW AT ZZ
2.25CAS Φ SCREW 8 OFF

Fig.8